

MATH 1564 — Notes 10/01

Markov chains — library example

A small town has two libraries, A and B .

Each month, among books checked out from library A :

- 80% are returned to A ;
- 20% are returned to B .

Each month, among books checked out from library B :

- 30% are returned to A ;
- 70% are returned to B .

If initially A has 500 books and B has 500 books, how many books will each library have after k months?

Model. Let

$$a_k = \text{number of books in } A \text{ after } k \text{ months}, \quad b_k = \text{number of books in } B \text{ after } k \text{ months}.$$

Then

$$a_{k+1} = 0.8a_k + 0.3b_k, \quad b_{k+1} = 0.2a_k + 0.7b_k.$$

In matrix form,

$$X_k = \begin{bmatrix} a_k \\ b_k \end{bmatrix}, \quad X_{k+1} = \begin{bmatrix} a_{k+1} \\ b_{k+1} \end{bmatrix},$$

and

$$X_{k+1} = PX_k, \quad P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}.$$

Initially,

$$X_0 = \begin{bmatrix} 500 \\ 500 \end{bmatrix}, \quad X_1 = PX_0, \quad X_2 = PX_1 = P^2X_0.$$

It is better to replace

$$\begin{bmatrix} a_k \\ b_k \end{bmatrix}$$

by

$$\frac{1}{1000} \begin{bmatrix} a_k \\ b_k \end{bmatrix}.$$

Then

$$\frac{a_k}{1000} = \text{the proportion of books in } A \text{ after } k \text{ months,}$$

and

$$\frac{b_k}{1000} = \text{the proportion of books in } B \text{ after } k \text{ months.}$$

Definition. A **probability vector** is a vector

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

such that every $x_i \geq 0$ and

$$x_1 + \cdots + x_n = 1.$$

Its coordinates are probabilities of possible outcomes.

Example. A coin toss has two outcomes, head and tail, represented by the probability vector

$$\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}.$$

Definition. A **stochastic matrix** is an $n \times n$ matrix whose columns are probability vectors.

Definition. A **discrete Markov chain** is a sequence of probability vectors in which each vector is determined by the preceding vector.

Example. If X_0 is a probability vector in $\mathbb{R}^n$ and P is a stochastic matrix, then

$$X_k = P^k X_0$$

is a probability vector for every k . Therefore, (X_k) is a Markov chain.

Theorem. If X_0 is a probability vector and P is stochastic, then

$$X_k = P^k X_0$$

is a probability vector for every k .

Proof. Use induction on k .

Base case: For $k = 0$, the vector X_0 is a probability vector by assumption.

Inductive step: Suppose X is a probability vector. We show that PX is a probability vector. Write

$$P = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

Then

$$PX = \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n \end{bmatrix}.$$

Every entry of PX is nonnegative because the entries of P and X are nonnegative. The sum of the entries of PX is

$$(a_{11} + \cdots + a_{n1})x_1 + \cdots + (a_{1n} + \cdots + a_{nn})x_n.$$

Because each column of P is a probability vector,

$$a_{1j} + \cdots + a_{nj} = 1 \quad \text{for each } j.$$

Therefore,

$$(a_{11} + \cdots + a_{n1})x_1 + \cdots + (a_{1n} + \cdots + a_{nn})x_n = x_1 + \cdots + x_n = 1.$$

Thus PX is a probability vector.

Conclusion. The recursion

$$X_{k+1} = PX_k$$

models a change in outcome probabilities from time k to time $k + 1$.

Example.

$$P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

is stochastic, and

$$X_0 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

is a probability vector. Then

$$X_1 = \begin{bmatrix} 0.4 + 0.15 \\ 0.1 + 0.35 \end{bmatrix} = \begin{bmatrix} 0.55 \\ 0.45 \end{bmatrix}.$$

Definition. A **steady state** for $X_{k+1} = PX_k$ is a probability vector q satisfying

$$Pq = q.$$

Example: steady state for the library model.

$$P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}.$$

Let

$$q = \begin{bmatrix} x \\ y \end{bmatrix}.$$

The equation $Pq = q$ is equivalent to

$$(P - I)q = 0,$$

so

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.2 & 0.3 \\ 0.2 & -0.3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -0.2x + 0.3y \\ 0.2x - 0.3y \end{bmatrix}.$$

Thus $2x = 3y$. Since $x + y = 1$,

$$x = \frac{3}{5}, \quad y = \frac{2}{5}, \quad q = \begin{bmatrix} 3/5 \\ 2/5 \end{bmatrix}.$$

Examples: describing steady states

1. If

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

then $Pq = q$ for every probability vector q . Thus every probability vector is a steady state.

2. If

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

then

$$(P - I)q = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} q = 0.$$

Hence $q_1 = q_2$, and the steady state is

$$q = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}.$$

3. If

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix},$$

then

$$(P - I)q = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -y \end{bmatrix} = 0.$$

Thus $y = 0$, so the steady state is

$$q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Theorem. Every stochastic matrix has a steady state.

Proof for 2×2 matrices. Write

$$P = \begin{bmatrix} r & 1-s \\ 1-r & s \end{bmatrix}, \quad 0 \leq r, s \leq 1.$$

Then

$$P - I = \begin{bmatrix} r-1 & 1-s \\ 1-r & s-1 \end{bmatrix}.$$

For $q = \begin{bmatrix} x \\ y \end{bmatrix}$, the equation $(P - I)q = 0$ is

$$(r-1)x + (1-s)y = 0, \quad x + y = 1.$$

If $r + s \neq 2$, this gives

$$q = \begin{bmatrix} \frac{1-s}{2-r-s} \\ \frac{1-r}{2-r-s} \end{bmatrix}.$$

If $r + s = 2$, then $r = s = 1$, so $P = I$ and every probability vector is a steady state.

Definition. A stochastic matrix P is **regular** if P^k has strictly positive entries for some positive integer k .

Example. Is

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

regular? No, because

$$P^{2j} = I, \quad P^{2j+1} = P,$$

so every power has a zero entry.

Example. Is

$$P = \begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$$

regular? No. Every power of an upper-triangular matrix is upper triangular, so the lower-left entry remains zero.

Example. Let

$$P = \begin{bmatrix} 0.5 & 0.3 & 0.4 \\ 0.5 & 0.3 & 0.1 \\ 0 & 0.4 & 0.5 \end{bmatrix}.$$

Then

$$P^2 = \begin{bmatrix} 0.4 & 0.4 & 0.43 \\ 0.4 & 0.28 & 0.28 \\ 0.2 & 0.32 & 0.29 \end{bmatrix},$$

whose entries are all positive. Therefore, P is regular.

Definition. A steady state q for a stochastic matrix P is a **sink** if, for every probability vector X_0 , the sequence defined by

$$X_{k+1} = PX_k$$

converges to q .

Example. If P has a sink q , then q is the only steady state of P . Indeed, if q_1 is another steady state, then $P^k q_1 = q_1$ for every k ; convergence to q forces $q_1 = q$.

Example. Does

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

have a sink? No. With

$$X_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad q = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix},$$

the vectors $P^k X_0$ oscillate.

Theorem. Every regular stochastic matrix P has a sink q such that, for every probability vector X_0 ,

$$\lim_{k \rightarrow \infty} P^k X_0 = q.$$

Determinants

For a 2×2 matrix,

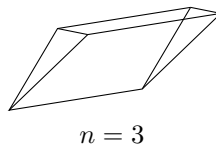
$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

For a 1×1 matrix,

$$\det[a] = a.$$

Motivation.

1. $\det : M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}$.
2. The determinant is easily computable.
3. $\det(AB) = \det(A) \det(B)$.
4. $\det A = 0$ if and only if A is not invertible (equivalently, A is singular).
5. $|\det[v_1 \ \cdots \ v_n]|$ is the volume of the parallelepiped spanned by $v_1, \dots, v_n$.



Definition: cofactor expansion along row one. If $A = (a_{ij})$ is $n \times n$, then

$$\det(A) = a_{11}C_{11} + \cdots + a_{1n}C_{1n},$$

where

$$C_{1j} = (-1)^{1+j} \det(A_{1j}),$$

and A_{1j} is the matrix obtained from A by removing row 1 and column j .

Example.

$$\begin{aligned} \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} &= aC_{11} + bC_{12} \\ &= a \det[d] + b(-\det[c]) \\ &= ad - bc. \end{aligned}$$

Example.

$$\begin{aligned} \det \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 2 \\ -1 & 0 & 1 \end{bmatrix} &= \det \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} - 2 \det \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} - \det \begin{bmatrix} 3 & 1 \\ -1 & 0 \end{bmatrix} \\ &= 1 - 2(3 + 2) - 1 \\ &= -10. \end{aligned}$$

Example. If A is upper triangular, then repeated cofactor expansion along the first row gives

$$\det(A) = a_{11}a_{22} \cdots a_{nn}.$$

Change in $\det A$ under row operations

Rules.

1. If B is obtained from A by swapping two rows, then

$$\det B = -\det A.$$

For a 2×2 matrix,

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc, \quad \det \begin{bmatrix} c & d \\ a & b \end{bmatrix} = cb - ad.$$

2. If B is obtained from A by scaling one row by a constant c , then

$$\det B = c \det A.$$

Intuitive check for row scaling.

$$\begin{aligned}\det \begin{bmatrix} a & b \\ 2c & 2d \end{bmatrix} &= -\det \begin{bmatrix} 2c & 2d \\ a & b \end{bmatrix} \\ &= -2 \det \begin{bmatrix} c & d \\ a & b \end{bmatrix} \\ &= 2 \det \begin{bmatrix} a & b \\ c & d \end{bmatrix}.\end{aligned}$$

Rule 3. Replacing a row by itself plus a multiple of another row does not change the determinant.

Intuitive check.

$$\begin{aligned}\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} &= ad - bc, \\ \det \begin{bmatrix} a & b \\ 2a + c & 3b + d \end{bmatrix} &= a(3b + d) - b(2a + c) \\ &= ad - bc.\end{aligned}$$